

L2A™

(b) Discuss the geographical distribution and natural selection of the ABO Histo-blood group systems

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ABO Blood Group is an example
of Multiple Allele Genetic Polymorphism.

Location: Chromosome 9

Apperance: 8th Month of Pregnancy

Discovery: By Karl Landsteiner in
1900.

Studies: → 1st Genetic Survey by
Morand in 1930;
→ McArthur & Penrose (1951)

Inheritance: Co-Dominance Pattern
Patten found by Bernstein (1924)

Alleles: I^A , I^B & I^O

Inheritance Pattern :

Each parent gives 1 allele to **L2ATM** form the zygote :

If

$I^A I^A$	}	$\rightarrow A$	}	Hence 4 <u>blood groups</u> possible
$I^A I^O$				
$I^B I^B$	}	$\rightarrow B$		
$I^B I^O$				
$I^O I^O$		$\rightarrow O$		
$I^A I^B$		$\rightarrow AB$		

Now if

$I^A I^A$	$\rightarrow$	'A' Antigen on RBC
$I^B I^B$	$\rightarrow$	'B' Antigen on RBC
$I^O I^O$	$\rightarrow$	No Antigen on RBC

So Only that Blood can be transfused if the victim doesn't have Antibody of the donor.

A $\rightarrow$ Accepts A, O
[Has Anti-B]

B $\rightarrow$ Accepts^{ss} B, O
(Has Anti-A)

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ATM

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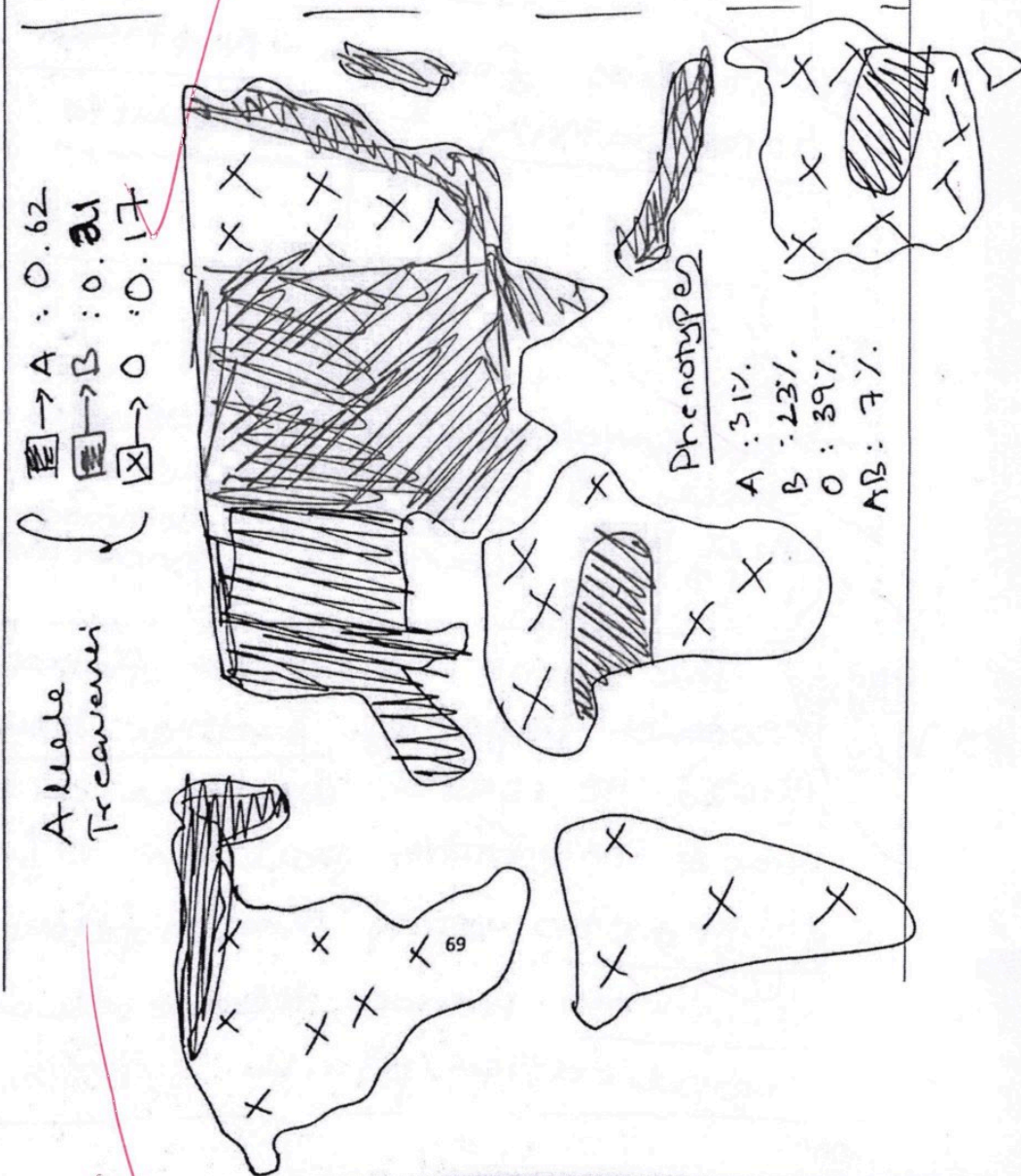
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AB (has No Antibody) → AB, B, A, O only L2ATM

O (has A&B Antibody) → Only O

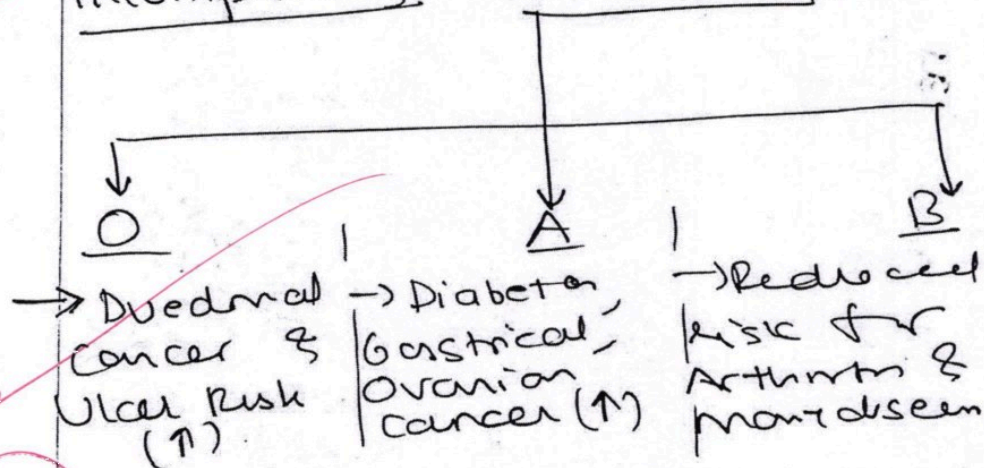
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∴ O is universal donor while AB is universal recipient



Does Natural selection Operate

1. Yes; as the frequencies **L2ATM** of A, B, O vary from population
2. Originated Bm79 and also found in Chimps, 6mllas, etc.
3. Studies focus on mother fetus incompatibility & pathogenesis.



12/11 ✓
The same was also found by research paper of Butter Janush (1962). He said we can also trace migration pattern of H. erectus using the population variation. Hence tremendous application possible.